ARDOR / INDEPENDENT FUNCTION MODULE

Stereo headphones / OPA1656 / relay mute

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J602 / HEADPHONE OUT
1: LEFT OUT 2: RIGHT OUT
3: GND 0V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

J102
STEREO AUDIO INPUT FROM SOURCE
C601
10u / 50V BIPOLAR
R611
20k / 1% 0.25W
U601A
OPA1656ID
VREF
R631
10R / 1% 0.25W
C651
470u / 16V
DRIVE_L
R651
10k / 1% 0.25W / UPRIGHT
C661
100p / 50V C0G
R641
1k / 1% 0.25W
GND
CHASSIS
D601
1.5KE6.8CA
HP_L
J602
STEREO HEADPHONE AUDIO OUTPUT
HP_L
HP_R
GND
C602
10u / 50V BIPOLAR
R612
20k / 1% 0.25W
U601B
OPA1656ID
VREF
R632
10R / 1% 0.25W
C652
470u / 16V
DRIVE_R
R652
10k / 1% 0.25W / UPRIGHT
C662
100p / 50V C0G
R642
1k / 1% 0.25W
GND
CHASSIS
D602
1.5KE6.8CA
K601
G5V-2-H1 DC5
HP_R
DRIVE_R
+5V
GND
GND
DRIVE_L
RELAY_LOW
HP_L
HP_R
J101
5V POWER + ENABLE INPUT
GND
+5V
R601
470R / 1% 0.25W
Q601
2N3904 / E-B-C
RELAY_LOW
GND
D603
1N4148
RELAY_LOW
+5V
R602
100k / 1% 0.25W
GND
+5V_A
R621
10k / 1% 0.25W
VREF
R101
0R / WIRE LINK
CHASSIS
GND
FB101
800R@100MHz / WE 7427501
+5V
+5V_A
C101
100u / 25V
C102
100n / 50V X7R
GND
GND
U601C
OPA1656ID
V+
V-
+5V_A
GND
+5V_A
C641
100n / 50V X7R / P2.5mm
R622
10k / 1% 0.25W
C631
47u / 25V
C632
100n / 50V X7R
GND
GND
GND
GND
+5V
+5V_A
CHASSIS
GND

Regulated 5V +/-5% / 150mA, 32-300 ohm headphones; source <=1Vrms.
OPA1656ID: one SOIC-8, 1.27mm pitch; all other components THT.
Inverting gain -10k/20k = -0.5; both inputs held at VREF +2.5V.
10R output isolation BEFORE 470u caps; + toward amplifier. Relay OFF grounds panel.
HP_ENABLE: 3.3V HIGH sources ~5mA; wait >=5s after stable supply/audio.
No onboard timer. Check DC/transients, oscillation, noise and clipping on dummy loads.

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_Headphones_THT.kicad_sch

Title: Stereo headphones / OPA1656 / relay mute

Size: A3	Date: 2026-10-12	**Rev: T1.1**
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